01 · Who should we email? — customer-level uplift (Anchor A)

July 13, 2026

The business decision. We have a discount offer and a customer list; contacting one customer costs **€8 all-in** (set with the simulation below — the bar a customer's uplift must clear). Emailing *everyone* burns margin on people who would have bought anyway, and a discount email can even annoy some loyal customers into unsubscribing. Emailing *no one* leaves money on the table. So the question is not "does the email work on average?" but **"for which individual customers does sending the email actually change what they do?"**

The one idea this whole notebook rests on

For each customer there are two possible futures: the spend we'd see **if we email them**, and the spend we'd see **if we don't**. Statisticians call these two futures *potential outcomes* and write them $Y(1)$ (emailed) and $Y(0)$ (not emailed). The thing we care about is the **difference** — how much the email *changes* that customer's spend:

$$\tau_i \;=\; Y_i(1) - Y_i(0). \qquad \text{(the customer's individual treatment effect, in €)}$$

We use the Greek letter τ **("tau")** for this effect throughout. When it depends on the customer's characteristics x (recency, engagement, …) we write it as a function $\tau(x)$.

The catch — the **"fundamental problem of causal inference"** — is that for any one customer we only ever get to see *one* of the two futures: we either emailed them or we didn't, never both. So τ_i is never directly observed for anybody. Everything that follows is a disciplined way of *filling in the future we didn't see.*

Four kinds of customer — and why only the effect tells them apart

Sort customers by what they do in each of the two futures, and four types appear. Notice that two customers can have the *same observed behaviour* ("bought after being emailed") yet be completely different types:

	buys if emailed	**doesn't buy if emailed**
buys if not emailed	🟠 *Sure Thing* — would buy anyway, so $\tau \approx 0$ (the discount is pure waste)	🔴 *Sleeping Dog* — the email **backfires**, $\tau < 0$ (reminding them prompts an unsubscribe/return)

	buys if emailed	**doesn't buy if emailed**
doesn't buy if not emailed	● *Persuadable* — the email *tips them over*, $\tau > 0$ (**the only type worth paying to reach**)	◦ *Lost Cause* — won't buy regardless, $\tau \approx 0$

Only the **Persuadables** are worth a discount. But you cannot identify them from who bought — a Sure Thing and a Persuadable both "bought after the email." You can only separate them by estimating the *effect* τ, which is why plain response models (predicting who buys) are the wrong tool and **uplift models** (predicting whose behaviour *changes*) are the right one.

Targeting is worthwhile **only because τ varies across customers** ("heterogeneity"). If every customer had the same τ, there'd be nothing to select on — you'd rationally email everyone (if τ > cost) or no one (if τ < cost). So the entire problem reduces to: **estimate the per-customer effect $\tau(x)$ well, then turn it into a euro decision that respects how unsure we are.**

How this notebook is organised

It follows the repo's fixed **7-step contract**, then adds **three depths** a data scientist needs to defend the number to a skeptic:

- **Steps** — 1 question · 2 simulate a known truth · 3 identify · 4 estimate · 5 validate · 6 decide in € · 7 caveats
- **Depth A** estimator bake-off & failure modes · **Depth B** identification rigour & sensitivity · **Depth C** euro policy, value-of-information & test design

> **A note on "Bayesian."** We fit models that return not a single number but a *distribution* of plausible values for each effect — a **posterior**. In the narration we never dwell on the machinery; a posterior just lets us say *"how sure are we, and does that change what we'd do?"* A **90% credible interval** below means "given the data and model, there's a 90% chance the true value lies in this range."

```
FAST=False  N=1600  sampling={'draws': 600, 'tune': 600, 'chains': 4, 'm': 100}
```

2 · Simulate a ground truth (why we start with fake data)

Here is the uncomfortable fact that shapes how we *validate* any causal method: **on real data we can never check whether the method got the effect right**, because the true effect τ_i is never observed for anyone (we only saw one of their two futures). If we just ran a fancy model on real data and it printed "€6.10", we'd have no way to know if that's right or nonsense.

The way out — used in every notebook here — is **validation-first**: we build a *simulator* that plants a **known** effect for each customer, generate data from it, and then see whether the method **recovers the number we planted**. Only after a method passes that test on data where we know the answer do we trust it on data where we don't. (Section 6b then runs the same pipeline on a real public dataset, where we can no longer grade recovery but can still make the decision.)

What the simulator plants. Each customer gets:

- five features x — `recency` (days since last purchase), `frequency`, `monetary`, `tenure`, `engage` (an engagement score);

- a true per-customer effect $\tau(x)$ (in €), deliberately *heterogeneous* — engaged, mid-recency customers are **Persuadables** (large positive τ); habitual high-frequency but low-engagement customers are **Sleeping Dogs** ($\tau < 0$, the email nudges them to unsubscribe); the rest sit near 0 (τ is continuous, so nobody is *exactly* zero — the Step-2 printout therefore shows only the two signed shares);
- a **selection rule** that decides who got emailed in the past. In the *observational* world we use here, marketers historically emailed the already-engaged, recently-active customers — so whether a customer was emailed is **tangled up with the kind of customer they are**. That tangling is called **confounding**, and it is exactly what makes the naive answer wrong (Step 4 and Depth A show it).

The data-generating model — exactly what `dgp.uplift_customers` implements (defaults & seed in `src/cmp/dgp.py`). Features: R = recency $\sim U(5, 330)$, F = frequency $\sim$ Poisson(5), M = monetary $\sim U(15, 180)$, tenure $\sim U(1, 60)$, E = engage $\sim$ Beta(2, 3). With $\sigma(z) = 1/(1 + e^{-z})$ the logistic function:

$$
\begin{aligned}
\mu_0(x) &= \max\Big\{0,\ 20 + 1.4\,F + 0.25\,M + 12\,E - 0.03\,R\Big\} \\
\tau(x) &= \underbrace{42\,e^{-\left(\frac{R-120}{70}\right)^2}\,\sigma(F-3)\,\sigma(9(E-0.28))}_{\text{Persuadables: engaged, mid-recency}} - \underbrace{24\,\sigma(9(0.22-E))\,\sigma(1.2(F-6))}_{\text{Sleeping Dogs: frequent but disengaged}} \\
e(x) &= \sigma\Big(1.8\,(E-0.4) + 1.2\,\tfrac{150-R}{150} + 0.15\,(F-5)\Big), \qquad T \sim \text{Bernoulli}(e(x)) \\
Y &= \mu_0(x) + \tau(x)\,T + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\,8^2).
\end{aligned}
$$

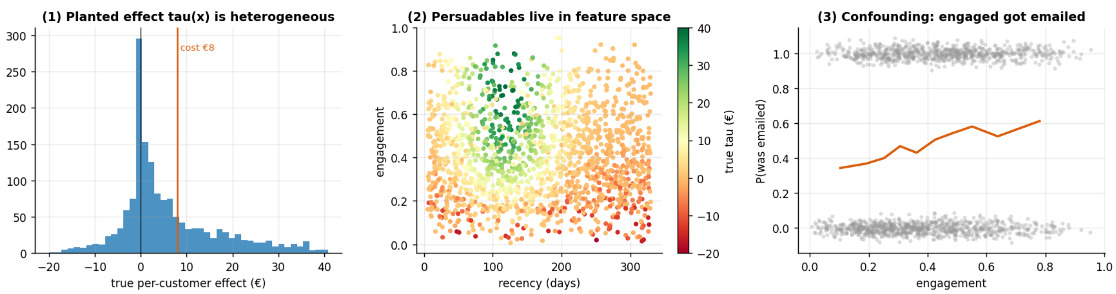
Line 1 is the baseline spend if not emailed; line 3 is the historical (observational) assignment rule. Three things to read off the equations. **(i) The confounding is visible**: engagement and recency raise the emailing propensity $e(x)$ *and* raise spend through $\mu_0(x)$ — that double role is what breaks the naive comparison. In the *randomized* regime the third line is replaced by $e(x) = \frac{1}{2}$ and the confounding vanishes. **(ii) `tenure` is a red herring**: it is drawn but enters neither μ_0 nor τ, so a good estimator must learn to ignore it. **(iii) Depth B's hidden confounder** adds an unobserved $U \sim \mathcal{N}(0, 1)$ to both the propensity logit ($+\,a_T U$) and the outcome ($+\,b_Y U$) — the two knobs the sensitivity analysis sweeps.

```
True ATE (average of tau over everyone) = €5.95
Share historically emailed              = 48%

Share of each uplift type in the base:
persuadable     0.7
sleeping dog    0.3
  (of the base, profitably persuadable at cost €8 - tau > cost: 30%)

      recency  frequency   monetary     tenure    engage    T          y  \
0  208.156027        4.0  84.060202  44.320568  0.545720  1.0  45.878520
1  296.594485        3.0  48.303531  32.765013  0.518337  1.0  32.005618
2  257.097849        5.0  46.447918  51.242745  0.921777  0.0  45.778013
3   78.192337        9.0  90.968991  39.685558  0.411379  0.0  48.419005
4  102.554043        4.0  44.561913  11.017535  0.447583  1.0  59.161710
```

```
         tau
0   5.658414
1  -0.008397
2   0.785915
3  18.905056
4  23.398746
```

The three panels below are the picture to hold in your head for the rest of the notebook:

1. **The planted effect is spread out** (not a single value) — that spread is the *heterogeneity* that makes targeting possible, and it straddles both 0 and the €8 cost line.
2. **Persuadables cluster in feature space** — high τ (green) lives at particular recency/engagement combinations, which is *why* a model using x can find them.
3. **The confounding, made visible** — the probability of having been emailed *rises with engagement*. Emailed and not-emailed customers are therefore **not comparable as-is**; that is the problem every method below has to overcome.



3 · Identify — what exactly are we estimating, and when is it even possible?

"Identification" vs "estimation" — the single most important distinction in causal inference. *Identification* asks: *given how the world works, can the effect in principle be recovered from this data at all?* That is a question about **assumptions**, answered before you fit anything. *Estimation* is the statistics you do *once identification is granted.* A beautiful model computed under a false identification assumption is **confidently wrong** — so we keep the two separate.

The estimand (the precise thing we want). We climb a ladder of three related quantities:

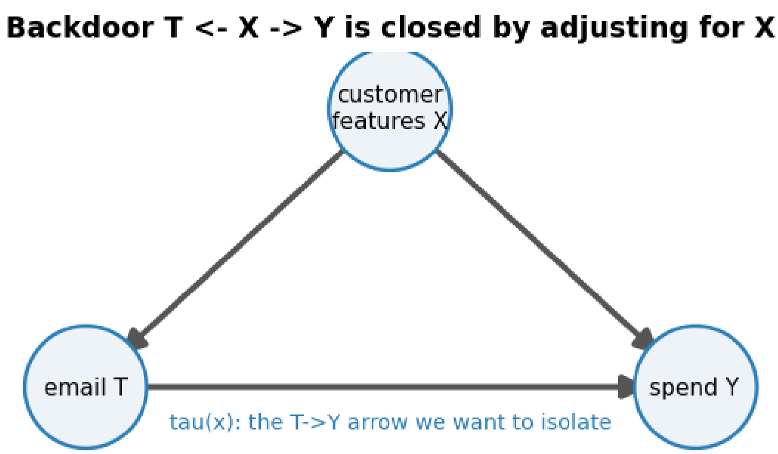
quantity	plain meaning	formula
ATE — Average Treatment Effect	"does emailing lift spend *overall*?"	$\mathbb{E}[Y(1) - Y(0)]$
CATE — Conditional ATE, $\tau(x)$	"the effect *for customers like x*" → *who* to target	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$
policy value	"the € we make from a given targeting rule"	$\mathbb{E}[(\tau(x) - c)\, \mathbb{1}\{\text{we target } x\}]$

($\mathbb{E}[\cdot]$ is just "average over customers"; $\mathbb{1}\{\cdot\}$ is 1 when the condition holds, else 0.) The uplift decision needs all three, but everything rests on the CATE $\tau(x) = \mu_1(x) - \mu_0(x)$, where $\mu_t(x) = \mathbb{E}[Y \mid T = t, X = x]$ is the *average spend of emailed (t=1) or not-emailed (t=0) customers who look like x.*

The three assumptions that make $\tau(x)$ recoverable (each is revisited in Depth B):

1. **Unconfoundedness** (a.k.a. *no unmeasured confounders*, "ignorability"): once we condition on the measured features X, who got emailed is as-good-as-random. Formally $\{Y(0), Y(1)\} \perp T \mid X$ ($\perp$ reads "is independent of"; $\mid X$ "once we hold the features fixed"). *Guaranteed by design in an A/B test; an assumption otherwise* — and the big one, because it can't be checked from data.
2. **Positivity / overlap**: every kind of customer had *some* chance of being emailed and *some* chance of not — $0 < e(x) < 1$, where $e(x) = P(T{=}1 \mid X{=}x)$ is the **propensity score** (a customer's probability of having been emailed). If some group was *always* emailed, we have no comparison for them and are extrapolating. Checked in Step 4.
3. **SUTVA**: one customer's email doesn't change another's outcome (no word-of-mouth spillover), and there's only one version of "the email."

Under 1–3, the **backdoor adjustment** (or *g-formula*) says $\tau(x) = \mu_1(x) - \mu_0(x)$: comparing emailed vs not-emailed *within* look-alike customers removes the confounding. The DAG (directed acyclic graph — a picture of what causes what) shows why. The "backdoor path" email $\leftarrow$ features $\rightarrow$ spend is the route by which confounding sneaks in; conditioning on the features **blocks** it.



Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst would do here **without** it. Not a different analysis: the *same estimands* of §3 — the ATE, then the CATE $\tau(x)$ — under the *same three assumptions* (unconfoundedness, overlap, SUTVA), estimated with the simplest tools that are still correct. No likelihood, no prior, no sampler. The causal work lives in the identification argument, not in the machinery, and this section proves it by doing the whole job without any.

The ATE, four ways — and the first one is a trap. The simplest estimator there is, the

difference in means ($\bar{y}_{\text{emailed}} - \bar{y}_{\text{not emailed}}$, with a Welch standard error that does not assume the two arms have equal variance), is *exactly right* — under randomization. Our main scenario is **observational** (§2's third line: the marketers emailed the engaged and the recently-active), so it is exactly wrong, and it is wrong *confidently*. To make the point unmissable we run it twice: on our confounded panel, and on a **randomized twin** — the same customers, the same planted $\tau(x)$, with the assignment rule replaced by a coin flip. Same estimator, same code, opposite verdicts. **What breaks is the identification, not the arithmetic.** Then we adjust, two ways:

- **OLS with the features** — $y \sim T + x$ — the workhorse regression adjustment. With a heteroskedastic cross-section (spend variance differs by arm and by customer type) the honest covariance is **HC1**, not iid: `cov="HC1"`.
- **OLS, Lin-interacted** — $y \sim T \times (x - \bar{x})$, i.e. *interact treatment with the centred features.* Because the covariates are centred, the coefficient on T is the ATE, and because each arm gets its own slopes, the estimator stays consistent for the ATE **even though the true** $\tau(x)$ **is heterogeneous** (Lin 2013). Plain additive OLS silently assumes one common effect, so with real heterogeneity it targets a variance-weighted average rather than the ATE — a subtle bias the interaction removes for the price of five coefficients.
- **AIPW** — the doubly-robust cross-check of §4, moved here where it belongs: it is a *frequentist* estimator with a bootstrap confidence interval and not one prior in sight. Cross-fitted (Chernozhukov et al.): the boosted-tree outcome models and the logistic propensity are fit on $K-1$ folds and scored out-of-fold, so an adaptive learner never grades its own homework. It is consistent if **either** the outcome model or the propensity model is right — see §4's influence function $\widehat{\psi}_i$ for the algebra.

All four are graded against the planted ATE, which we happen to know.

```
THE PLANTED TRUTH:  ATE = EUR 5.95   (all figures in EUR)

estimator                        estimate         90% CI covers truth?
ATE (difference in means)           11.37  [ 9.91, 12.83]           NO
ATE (OLS, additive covariates)       6.56  [ 5.62,  7.50]          YES
ATE (OLS, Lin-interacted)            6.39  [ 5.49,  7.29]          YES
ATE (AIPW, doubly-robust)            6.28  [ 5.39,  7.16]          YES

  …the SAME estimator on the RANDOMIZED twin (identical planted tau):
ATE (difference in means)            5.21  [ 3.79,  6.63]          YES

  standard errors, stated not defaulted: Welch (unequal variances) for the two
  differences in means; HC1 heteroskedasticity-robust for both OLS fits; cross-
fitted
  influence function + 500-resample percentile bootstrap for AIPW.

BIAS vs LUCK -- the SAME estimators over 12 fresh panels of each regime
(mean error against each panel's own planted ATE, in EUR):
  a mean error is BIAS when it is far from zero relative to ITS OWN standard
error,
  SE = sd/sqrt(12) -- NOT relative to the panel-to-panel sd, which measures
something else
  entirely (how much one draw jumps around). Both are printed; the tag uses the
```

```
SE.

  naive * observational                 +5.87  (sd 0.69, SE 0.20)  <- BIAS, 29.7
SEs from 0
  naive * RANDOMIZED                    +0.10  (sd 0.58, SE 0.17)  <- noise
around zero, 0.6 SEs from 0
  OLS Lin-interacted * observational    +0.44  (sd 0.40, SE 0.11)  <- BIAS, 3.8
SEs from 0

THE TABLE ANSWERS TWO SEPARATE QUESTIONS -- keep them apart.
(1) IS IT ZERO? The naive comparison is not unlucky, it is WRONG: off by roughly
the
    same +5.9 EUR every time (30 SEs from zero), while the IDENTICAL estimator
on
    RANDOMIZED panels sits a fraction of an SE from zero. Nothing changed but
HOW
    CUSTOMERS WERE SELECTED. The estimator was never the problem; the
identification was.
(2) HOW BIG IS WHAT SURVIVES? Adjustment does not drive the error to exactly
zero.
    What is left is +0.44 EUR -- 3.8 SEs above zero, so we call it what it is --
a small but REAL residual.
    Either way it is ~13x smaller than the naive arm's and ~7% of the 6.0 EUR
effect we
    are trying to measure. Adjustment REPAIRS the identification; it does not
ANOINT the
    estimate. 'Statistically distinguishable from zero' and 'big enough to
change a
    decision' are two different sentences, and this cookbook keeps them apart --
which is
    exactly why the tag above is computed against the SE OF THE MEAN and not
eyeballed
    against the panel-to-panel sd, a bar lax enough to wave a genuine 4-SE bias
through.
```

The ATE is not the decision. §1 said it plainly — "does emailing work on average?" is the wrong question, because a positive average can hide the Sleeping Dogs the email actively costs us. The business needs $\tau(x)$, per customer. So: **can the classical arm do heterogeneous effects?**

Yes, and it is not exotic. The **T-learner** ("T" = two) is three lines of scikit-learn: fit one regression of spend on features among the **emailed**, another among the **not emailed**, and subtract their predictions at every customer's x.

$$\hat{\tau}(x) \;=\; \hat{\mu}_1(x) \;-\; \hat{\mu}_0(x), \qquad \hat{\mu}_t \;=\; \text{a regression of } y \text{ on } x \text{ fitted on the arm } T = t.$$

That is the backdoor adjustment of §3, done twice. Under the *same* three assumptions it estimates the *same* $\tau(x)$ the Bayesian model will. No prior, no posterior, no MCMC — and a per-customer number we can rank, threshold and score on the notebook's **own** metrics (PEHE, correlation, AUUC/Qini — all four are defined formally in **§5**; read the printouts below as *lower PEHE is better, higher correlation and AUUC are better, and AUUC is scaled so 1.0 = the oracle's ranking and 0 = random*). We fit three versions to separate the *idea* (two arms, subtract) from the *base learner*:

- **T-learner (GBM)** — gradient-boosted trees per arm; the honest classical default, and flexible enough for the DGP's Gaussian-bump-times-logistic $\tau(x)$.
- **T-learner (OLS)** — the same recipe with a *linear* model per arm. It is the control experiment for functional form: $\tau(x)$ is planted as a smooth non-linear ridge in (recency, engagement) space, and a linear model cannot bend to it.
- **S-learner (GBM)** — Depth A's labelled failure mode, in classical clothing: *one* model with treatment as an extra column, so the regulariser can shrink the treatment's influence toward zero and **flatten** the heterogeneity. Included so the comparison is complete and the failure is *measured* rather than asserted.

The interval is where the classical arm runs out of road — and we measure exactly where. A T-learner returns a *point* $\hat{\tau}(x)$ and nothing else: no per-customer interval falls out of it. The classical escape hatch is the **nonparametric bootstrap** — resample the customers, refit both arms, repeat — which does give a band per customer. We compute it (it is cheap) and then grade it honestly against the truth: how often does the 90% bootstrap interval actually contain that customer's planted τ_i, overall and in the deciles where the money is? Keep one thing in view while reading the answer: a bootstrap band describes the **sampling variability of the estimator** — where $\hat{\tau}(x)$ would land if we redrew the customer list — and *not* a probability distribution for that customer's effect. §6's rule needs $P(\tau_i > \text{cost})$. Whether the bootstrap can stand in for it is a claim we test, in euros, in 5x.

```
CLASSICAL CATE -- graded on THIS notebook's own metrics (cmp.metrics), vs the
planted tau(x).  PEHE = RMSE against the true tau, in EUR (lower is better);
AUUC: 1.0 = the oracle's ranking, 0 = no better than random.
(All four metrics are defined formally in section 5; they are used here first.)

estimator                   ATE    PEHE   corr   AUUC
T-learner (GBM)            6.52    5.21   0.89   0.89
S-learner (GBM)            6.11    6.44   0.85   0.89
T-learner (OLS)            6.39    8.79   0.56   0.62

  -> No Bayes anywhere, and the GBM T-learner already ranks customers at AUUC
0.89 of
     the oracle. Functional form, not philosophy, is what costs you: the same
recipe
     with a LINEAR learner per arm drops to AUUC 0.62.
     The S-learner flattens tau(x) -- its shared treatment column is regularised
toward
     zero -- and the flattening costs it MAGNITUDES (PEHE 6.44 vs 5.21) while
costing
     it, with THIS base learner, essentially nothing on the RANKING (AUUC 0.89
vs 0.89):
     a uniformly squashed tau(x) can still put customers in the right ORDER.
Depth A
```

```
     repeats the bake-off with BART base learners and prints the S-vs-T verdict
axis by
     axis; read it there rather than here. The failure mode is real, but WHICH
axis it
     breaks -- magnitudes, ranking, intervals -- is a property of the base
learner, not
     of the letter 'S'.


BOOTSTRAP INTERVALS for the T-learner (200 resamples), graded against the
planted tau_i:
  overall 90% coverage 86% (nominal 90%)   *   mean width EUR 13.0
  TOP decile of true effect (where the money is): 78%   *   BOTTOM decile
(sleeping dogs): 78%
  -> broadly honest in the middle, fraying at BOTH extremes -- exactly where a
     targeting rule lives. And it is the wrong object: a band around the
ESTIMATOR,
     not a belief about tau_i.

What this 90% confidence interval does NOT say: that there is a 90% probability
the true effect lies inside it. It is a property of the *procedure* - 90% of
intervals built this way would cover the truth across repeated samples. This
interval either contains the truth or it does not. To get a probability *about
the effect itself* - the thing a go/no-go rule like P(lift > cost) >= 0.9
actually needs - you need a posterior. That is what the Bayesian section adds.
```

Read-out — the classical answer, in business terms. Printed above, in order:

1. **The naive comparison is a trap, and the trap is identification.** "Emailed customers spent €11-ish more" nearly *doubles* the true €6-ish effect, with a tight interval that **misses the truth entirely** — the textbook shape of *precisely wrong*. And the **bias-vs-luck** block settles that this is not one unlucky draw: averaged over fresh panels, the naive estimator is off by roughly the same amount *every time* (the print states it; it is close to the full size of the true effect), while the **identical estimator on randomized panels scatters around zero**. (Any single randomized twin can still miss its interval by chance — that is what a 90% interval means — which is exactly why we grade the *mean error over panels*, not one draw.) Nothing changed but who got picked: the failure is in the identification, not the arithmetic. Note the tag rule the print uses: a mean error counts as **bias** when it is more than two **standard errors of the mean** (sd/$\sqrt{n_{\text{panels}}}$) from zero — *not* two panel-to-panel sds, which is a different and much laxer bar (lax enough to wave a genuine 4-SE bias through as "noise"). That distinction *is* this cell, so the code applies it and prints how many SEs from zero each arm sits. The adjusted arm's small residual error is reported *with its own SE* rather than waved away, and whether it clears the bar is the print's call on the run you are looking at, not this paragraph's.
2. **Adjustment repairs it, and the three adjusted estimators agree.** Additive OLS, Lin-interacted OLS and cross-fitted AIPW land within a euro of each other and of the planted ATE, each with the truth inside its 90% interval. Three different bias-correction strategies (regression, regression with arm-specific slopes, doubly-robust) converging on one number is a *credibility* signal, and we bank it before any prior is written down.

3. **Heterogeneous effects do not require Bayes.** The classical T-learner delivers a per-customer $\hat{\tau}(x)$ that ranks customers at the AUUC printed above — a large fraction of the oracle's ranking, from two scikit-learn fits and a subtraction. Anyone who tells you that you *need* a posterior to do uplift is selling something. What you need is (i) the identification argument of §3 and (ii) a base learner flexible enough for the true $\tau(x)$ — and the OLS row shows that the second is not free: the same recipe with a linear learner collapses.

This is the honest baseline the Bayesian section must now beat. Not a strawman: a competent, correctly-standard-errored, doubly-robust-cross-checked, oracle-graded classical analysis, which has already produced (a) the right ATE, (b) a per-customer effect, and (c) a defensible ranking. Section 5x will hold §4's posterior to exactly this bar and report what it does and does not buy.

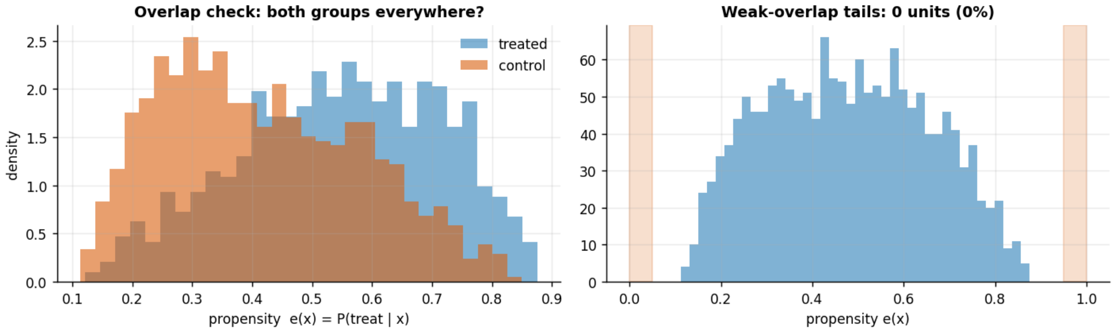
The guardrail — what those confidence intervals do *not* say (printed above, in the words this cookbook uses in every notebook): a 90% CI is **not** a 90% probability that the true effect is inside it. It is a property of the *procedure* — 90% of intervals built this way cover the truth across repeated samples — and this one either covers it or it does not. The same applies, exactly, to the bootstrap band we just put around each customer's $\hat{\tau}(x)$: resampling the customer list approximates *the sampling distribution of the T-learner*, so the band is centred on $\hat{\tau}(x)$, not on τ_i, and every replicate refits the same learner on the same information — a systematically shrunken estimate stays shrunken in all 200 of them, and the band tightens **around the wrong value** without a hint of complaint. That is visible in the printout: coverage sags at **both** extremes of the true effect — the top and bottom deciles print side by side above, and it is the *pair* that is the finding, not a ranking between them. (BCF's top decile is a different story, and a worse one; 5x puts the two arms' decile coverage next to each other rather than letting either cell speak for the other.)

And §6's rule is $P(\tau_i > \text{cost}) \geq 0.8$ — a probability **about this customer's effect**, one number per customer, feeding a threshold that a CMO can dial. The classical arm has no such object. It can offer the bootstrap *exceedance share* — the fraction of resamples in which $\hat{\tau}^*(x)$ happened to exceed €8 — which looks like a probability, is not one, and carries no guarantee of behaving like one. Whether that distinction costs real money, or is a philosopher's quibble, is not something we will assert. **5x measures it, in euros.**

4 · Estimate — check overlap first, then fit two independent ways

Discipline: look at overlap *before* estimating. Adjustment can only compare emailed vs not-emailed customers where *both* actually exist. If, at some feature combination, everyone was emailed, there is no control to compare against and the model is silently extrapolating. So we first plot the **propensity score** $e(x)$ by arm (emailed vs not); healthy overlap means the two histograms cover the same range, with few customers stuck at 0 or 1.

```
Overlap is adequate - only 0 of 1600 customers fall in the <0.05 / >0.95 tails.
```



Now the estimators. We fit the effect two completely different ways and check they agree — agreement across independent methods is our first credibility signal.

- **BCF — Bayesian Causal Forest** (our primary CATE model). Under the hood it's **BART** (Bayesian Additive Regression Trees — a flexible sum-of-trees model that needs no functional-form guesswork and returns a posterior). BCF is a causal-tuned BART with two tricks: it feeds the estimated propensity $e(x)$ into the model so *targeted-selection confounding* is absorbed (**targeted selection** is our data's specific confounding pattern: marketers emailed customers *because* they looked likely to spend, so the propensity $e(x)$ and the baseline spend $\mu_0(x)$ move together — feeding $\hat{e}(x)$ in lets the model soak up exactly that overlap, which is why BCF beats a plain T-learner precisely here, and only here, per Step 7's caveat), and it puts **fewer trees on the treatment surface** than on the baseline surface (a weak, tree-count form of BCF's shrinkage-toward-homogeneity prior), so the model only reports heterogeneity when the data really support it (rather than reading noise as signal). Output: a posterior array of shape `(draws, customers)` — one plausible $\tau(x)$ curve per posterior draw.
- **AIPW — Augmented Inverse-Propensity Weighting** (a cross-check on the *average*) — **already fitted, back in Step 0**, because it is a purely classical estimator: no prior, no sampler, a bootstrap confidence interval. We do not refit it here; we simply hold BCF's posterior-mean ATE up against it. It's **doubly-robust**: it combines an outcome model (gradient-boosted trees here) and a propensity model, and is correct if *either one* is right. It's a **methodologically different** cross-check (a different outcome model plus explicit propensity weighting), so agreement is reassuring though not fully independent.

The fitted model (BCF), in symbols. With $\hat{e}(x)$ the propensity estimated above:

$$y_i \sim \mathcal{N}(\mu_i,\, \sigma^2), \qquad \mu_i = f_{\text{prog}}(x_i,\, \hat{e}(x_i)) \;+\; f_\tau(x_i)\, T_i, \qquad \sigma \sim \text{HalfNormal}(15),$$
$$f_{\text{prog}} \sim \text{BART}(m \text{ trees}), \qquad f_\tau \sim \text{BART}(m/2 \text{ trees}),$$

where f_{prog} is the prognostic surface (baseline spend as a flexible function of the features *and* the propensity — feeding $\hat{e}$ in is the trick that absorbs targeted-selection confounding) and f_τ is the treatment surface, whose posterior **is** the CATE: $\hat{\tau}(x) = f_\tau(x)$. Halving the tree count on f_τ is the shrink-toward-homogeneity prior from the bullet above, in its concrete form. (The HalfNormal(15) on σ is just a mild prior keeping the noise sd positive and around the tens-of-euros scale — nothing hinges on the exact 15.)

The cross-check (AIPW), in symbols. AIPW averages one *influence-function* score $\hat{\psi}_i$ per customer — think of it as each customer's individual contribution to the average effect, engineered so that errors in either model wash out (see below) — built from an outcome model $\hat{\mu}_t(x)$ (predicted spend under treatment $t \in \{0, 1\}$) and the propensity $\hat{e}(x)$:

$$\hat{\psi}_i \;=\; \underbrace{\hat{\mu}_1(x_i) - \hat{\mu}_0(x_i)}_{\text{outcome-model guess}} \;+\; \underbrace{\frac{T_i\,(y_i - \hat{\mu}_1(x_i))}{\hat{e}(x_i)} \;-\; \frac{(1-T_i)\,(y_i - \hat{\mu}_0(x_i))}{1-\hat{e}(x_i)}}_{\text{inverse-propensity-weighted residual corrections}}, \qquad \widehat{\text{ATE}} \;=\; \frac{1}{n}\sum_{i=1}^{n} \hat{\psi}_i.$$

This equation is *why* "doubly robust" is more than a slogan: if the outcome models are right, the residuals $y_i - \hat{\mu}_{T_i}(x_i)$ are mean-zero and the correction terms vanish on average, leaving the correct first term; if instead the propensity is right, the inverse-weighted residuals exactly repair whatever bias the outcome models leave behind. Either model correct suffices — only both wrong breaks the estimator.

The pymc-bart gotcha, baked in. Scoring BART counterfactuals ("what would this customer have spent under the *other* treatment?") silently returns *frozen training predictions* — every effect exactly 0 — unless the tree node is resampled. `cmp.estimators` always does this correctly, and a unit test asserts non-zero recovered effects, so you never hit the trap.

Did the sampler converge? (the check behind every posterior in this cookbook). All the Bayesian fits here draw posterior samples with **MCMC** (Markov-chain Monte Carlo — the standard algorithm for drawing from a posterior), and a posterior is only trustworthy if the sampler actually converged. Three standard diagnostics, in plain terms:

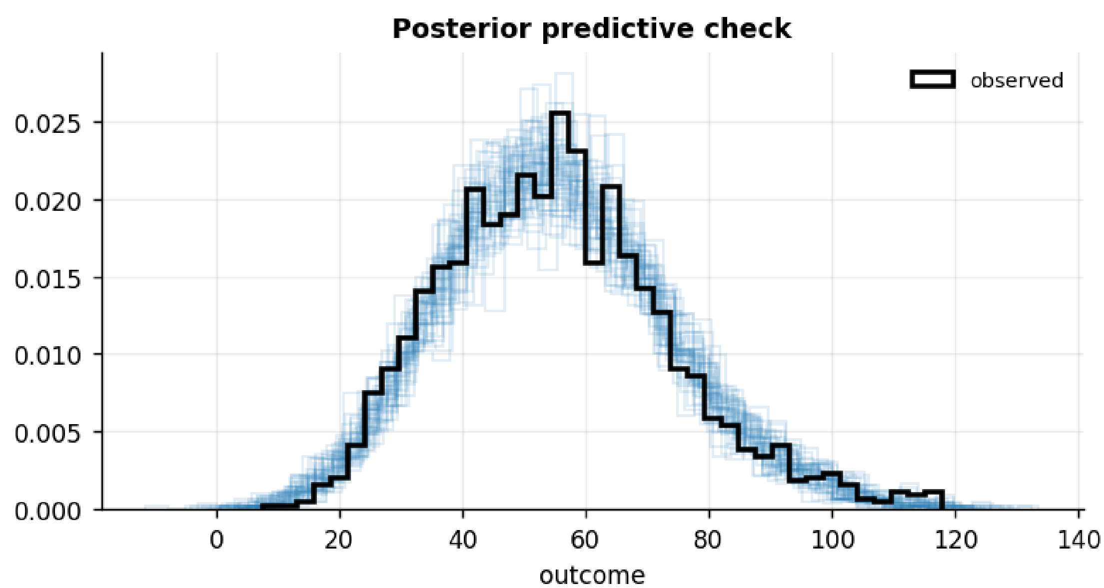
- **R-hat** — run several independent chains and compare them: if they disagree about where the posterior is, they haven't converged. R-hat ≈ 1.00 is good; above ~1.01 means keep sampling (or fix the model). One honest exception, applied in the very next cell: BART is fitted with the particle-Gibbs **PGBART** sampler, and scalars coupled to its tree sums (like the observation-noise sd we monitor) routinely sit at r-hat ≈ 1.01–1.02 in FULL runs even when the fit is fine — and, unlike pure NUTS, raising draws grinds that down only slowly. So for the BART-based fits we read ≤ **1.01 as a clean pass, 1.01–1.02 as *tolerated*** (with the 8-seed stability sweep in Step 5 as the backstop that the *estimates* are stable), **and > 1.02 as a genuine investigate signal**. The printout below applies exactly these bands, so the verdict you see is computed, not asserted.
- **ESS** (*effective sample size*) — consecutive draws are autocorrelated, so 1,000 draws may carry the information of only 100 independent ones. ESS is that independent-equivalent count; you want it comfortably in the hundreds for stable interval endpoints.
- **Divergences** — a NUTS-specific warning that the sampler hit posterior geometry it couldn't integrate; more than a handful means the posterior is suspect (raise `target_accept` or reparameterize). BART is fitted with a particle-Gibbs sampler instead of NUTS, so divergences don't apply here — but R-hat and ESS still do.

PyMC and CausalPy print these warnings automatically when something is off — silence is the pass signal. We keep them out of the notebook read-outs to stay focused on the causal logic, with one exception worth knowing about: notebook 06's MMM is hard enough geometry that FULL runs need `target_accept = 0.99` and deeper trees to reach R-hat ≈ 1.01.

```
BCF   ATE = €6.37   (true €5.95)
BCF   sampling convergence (4 chains): max r-hat 1.010 - min ESS 620 -
divergences 0
      r-hat verdict: PASS - max r-hat 1.010 is at or under the ~1.01 bar
AIPW  ATE = €6.28   90% CI [5.39, 7.16]   (doubly-robust, from Step 0)
BCF's ATE lies within the AIPW 90% CI - two methodologically different
estimators agree, our first credibility check.
```

Model criticism — the posterior predictive check (PPC). Before we *read effects off* a model, we ask whether it can even reproduce the data it was trained on. We draw fake ("replicated") datasets from the fitted model and overlay their spend distribution on the real one. If the real data looked nothing like what the model can generate (wrong spread, missed skew), the effect estimates would be untrustworthy no matter how tight their intervals.

```
Posterior-predictive check: 90% of observed spend falls within the replicates'
5-95% range (~90% if calibrated) - no gross misfit (posterior sigma ~ €9).
```



5 · Validate — did we recover the truth, is the uncertainty honest, and does it rank well?

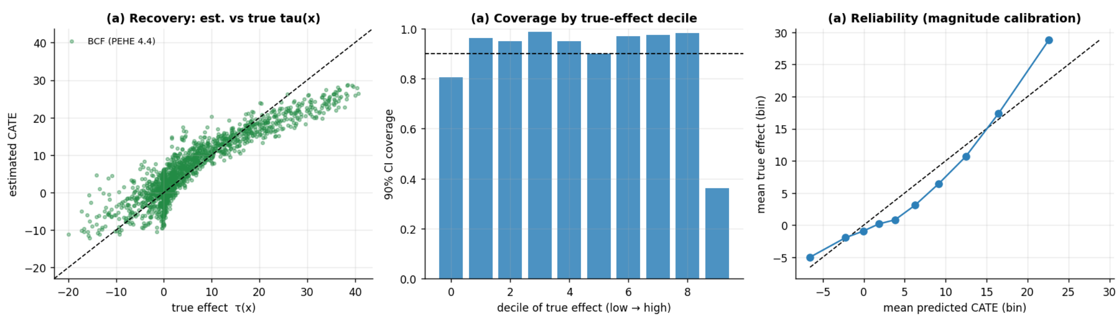
This is the payoff of simulating a known truth. Three questions, each with its own diagnostic:

(a) Recovery & calibration — are the numbers right, and honestly uncertain? - PEHE (*Precision in Estimation of Heterogeneous Effects*) — the root-mean-square error between the estimated $\hat{\tau}(x)$ and the true $\tau(x)$, in €. Lower is better; it's the CATE analogue of RMSE. - **Coverage** — a *90% credible interval* should contain the true effect about 90% of the time. We also break coverage out *by decile of the true effect* to catch a model that's well-calibrated on average but overconfident for the biggest effects. - **Reliability** — bucket customers by predicted effect and plot mean-predicted vs mean-true; on the 45° line means the *magnitudes* are calibrated, not just the ranking. - **Sharpness** — the average width of the intervals. Sharp *and* calibrated is the goal; sharp *and* miscalibrated is just overconfidence, which is why we never report sharpness alone.

(b) Ranking quality — for targeting, the *order* matters more than exact magnitudes. - **Qini / uplift curve** — walk down customers from highest to lowest predicted effect, contacting as you go, and plot the cumulative *true* effect captured. A good model's curve bows up toward the **oracle** (a hypothetical model that knows every true τ) and well above the **random** diagonal. - **AUUC** (*Area Under the Uplift Curve*) — that curve's area, normalised so **1.0 = oracle-perfect ranking** and **0 = no better than random**. The single number for "how well do we rank?" - **Uplift-by-decile** — average *true* effect within each tenth of predicted effect; a good model shows a clean descending staircase (top decile = biggest real effect).

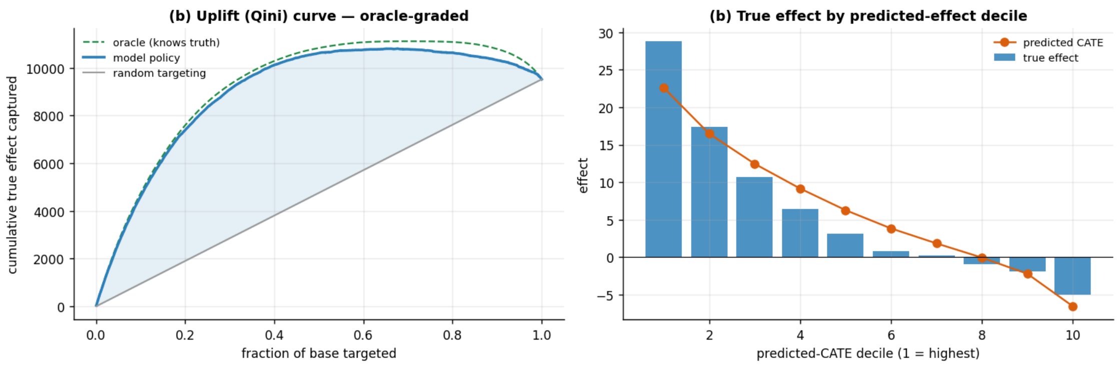
(c) Stability — is the result a fluke of one random draw? We refit on several fresh simulations and check the recovery is consistent.

```
PEHE €4.45  *  corr(est,true) 0.92  *  90% coverage 88%  *  interval sharpness
€13.3
```



Read-out. The recovery scatter and the line just printed tell one story: a high est-vs-true correlation and a small **PEHE** (a few € against true effects that span roughly −€10 to +€40) say the *ranking* is strong and the magnitudes are close, while the printed **90% coverage** lands a touch under its 90% target — a mild over-confidence that the middle panel pins to the largest-effect deciles and that Step 5's coverage caveat (below) returns to. Sharpness is read only alongside that coverage, never on its own, so a narrow-but-miscalibrated interval can't pass itself off as precision.

```
AUUC = 0.94 of oracle (1.0 = perfect ranking, 0 = random).
(oracle-graded: uses the known tau; on real RCT data use metrics.qini_observed,
which reweights
 observed outcomes by arm and needs no counterfactual - see §6b.)
Top decile true effect €28.8  vs  bottom decile €-5.0 - the model concentrates
the real uplift where we'd actually spend.
```



Heads-up on the output below: the sweep deliberately refits BCF with **short 80-draw chains** — enough for stable posterior *means* (all we read here), not for interval-grade diagnostics. The repeated `Only 80 samples per chain` notices are that choice talking; we leave them visible rather than model the habit of silencing sampler diagnostics on a cell whose output we quote.

```
(c) Stability across fresh simulations:
 seed  ATE_hat  true_ATE  PEHE  AUUC
  100     6.00      5.61  5.08  0.93
  101     5.98      6.12  4.87  0.94
  102     5.79      5.64  5.23  0.94
  103     6.26      5.63  5.05  0.93
  104     5.98      5.75  5.19  0.94
  105     5.95      5.78  5.45  0.92
  106     5.20      5.34  4.75  0.94
  107     6.09      5.84  5.14  0.94

ATE recovery bias across seeds: +0.19 (sd 0.25) - consistent, not a lucky seed.
```

5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0 built a complete classical uplift model with no priors in it: a per-customer $\hat{\tau}(x)$ from a GBM T-learner, plus a bootstrap band. §4 built a BCF posterior. **Same estimand, same three assumptions, same data.** Below they meet on this notebook's own four axes — magnitude (PEHE), ranking (AUUC/Qini), interval honesty (coverage, overall *and* by decile of the true effect), and sharpness — and then on the only grade that pays anyone: the **euro decision** of §6, run early here so the comparison is complete.

The decision rules put a fair, *unhelpful-to-us* version of the classical arm on the field. Rather than score it only on "point estimate > cost", we also let it use the bootstrap: a lower-confidence-bound rule (*is the whole 90% band above the cost?*) and — the closest thing the frequentist toolbox has to §6's rule — the **bootstrap exceedance share**, the fraction of resamples in which $\hat{\tau}^*(x) >$ €8, thresholded at the same 0.8. If the posterior's advantage is real, it has to show up against *those*.

```
RECOVERY, RANKING, CALIBRATION -- the four axes of section 5, both arms:
arm                                PEHE   corr   AUUC   cov90  top-dec  bot-dec
width
```

```
classical T-learner (Step 0)   5.21   0.89   0.89     86%      78%      78%
13.0
BCF posterior (section 4)      4.45   0.92   0.94     88%      36%      81%
13.3


  WINNER -- classical vs BCF, with a tolerance. A gap inside 2x its own paired-
bootstrap
  SE prints as a DEAD HEAT, because that is what it is:
    PEHE    BCF              (5.21 vs 4.45;  |gap| 0.76 >  2xSE = 0.24)
    AUUC    BCF              (0.894 vs 0.941;  |gap| 0.047 >  2xSE = 0.010)
    cov90   BCF              (0.039 vs 0.016;  |gap| 0.023 >  2xSE = 0.021)
[distance from the nominal 0.90: 0.86 vs 0.88]
    top-dec classical        (0.78 vs 0.36;  |gap| 0.42 >  2xSE = 0.09)
[coverage on the top true-effect decile]
  Depth A prices WHY, by putting a BART T-learner -- posterior averaging, but
none of BCF's
  extra machinery -- on the same data, and printing the attribution.

THE EURO DECISION at cost EUR 8 -- realised profit on the KNOWN truth:
  targeting rule                                   contacted      profit  % of
oracle
  email everyone                                        1600    -3,277 EUR
-60%
  classical * point CATE > cost                          589     4,815 EUR
88%
  classical * bootstrap 90% band all > cost              346     4,891 EUR
90%
  classical * bootstrap share > 0.8                      449     5,137 EUR
94%
  BCF * posterior mean > cost                            621     5,058 EUR
93%
  BCF * P(tau > cost) > 0.8   [section 6's rule]         444     5,210 EUR
95%
  oracle (knows every true tau)                          473     5,459 EUR
100%

  the posterior rule earns EUR +72 against the BEST classical rule on
  the board -- 1.3% of the EUR 5,459 oracle prize. Whatever the Bayesian
  layer is buying in this notebook, on THIS decision, at THIS n, it is not a
  pile of euros. Say it out loud.

AND THE COLUMN THAT DOES NOT EXIST. Both arms hand section 6 a number in [0,1]
per
customer. Bucket customers by it and ask: of those told '~p', what SHARE truly
had
tau_i > cost?
  bucket             n  BCF: claim   actual    |     n  boot: claim   actual
```

```
(0.0, 0.2]       773         0.05      0.00     |   822         0.04     0.00
(0.2, 0.4]       145         0.29      0.00     |   166         0.29     0.11
(0.4, 0.6]       117         0.50      0.12     |    86         0.50     0.20
(0.6, 0.8]       121         0.71      0.44     |    77         0.71     0.40
(0.8, 1.0]       444         0.96      0.91     |   449         0.97     0.90

  Read each 'actual' column against its 'claim'. Neither number is a perfectly
  calibrated probability -- both are optimistic in the middle buckets. But only
ONE
  of them is even TRYING to be one: P(tau_i > cost) is a statement about the
  CUSTOMER, and it is what section 6's rule, the confidence sweep, the straddler
set
  and the euro value-of-information in Depth C are every one of them written in.
  The bootstrap share is a statement about the ESTIMATOR -- how often a refitted
  T-learner would land above EUR 8 if we redrew the customer list. It is centred
on
  tau-hat, not on tau_i, and nothing licenses reading it as a belief about a
person.
```

The honest verdict — what the posterior bought, and what it did not.

1 · On the ATE, they agree — say it plainly. Step 0's adjusted classical estimators and §4's BCF posterior land on the same ~€6 average effect, all of them within a euro of the planted truth. The Bayesian layer did not find a better number. It could not: with 1,600 customers, fully observed confounders and weak priors, a correctly-adjusted regression is already consistent, and there was nothing left on the table. **The causal work was done by the identification of §3, not by the machinery of §4.** If you came here expecting "going Bayesian" to fix a biased estimate, Step 0 is the correction: what fixed the bias was *adjusting for the five features*, and OLS does that for free.

2 · On the CATE the Bayesian arm wins — but *which ingredient* won is a separate question, and we make the code answer it. Read the `WINNER` lines under the first table (never this paragraph, if the two ever disagree). They are **tolerance-guarded**: a gap smaller than twice its own paired-bootstrap standard error prints as *too close to call*, because that is what it is. On top of that sampling noise, `pymc-bart` is **not seed-reproducible in this environment**, so every BART number in this notebook moves between runs and the printed tolerance is a *floor* on the wobble, not a ceiling. **Nothing here is "seed-stable", and we no longer claim it is.** Hold the size of any BART gap loosely, in either direction — an earlier draft of this notebook was caught running a **double standard** on exactly that: it dismissed the *larger* of two BART PEHE gaps as "noise" where the gap pointed *away* from Bayes, and banked the *smaller* one as "a real, measured Bayesian win" where it pointed *toward* it. Neither gap was stable. One standard, applied by code, to every comparison — that is what the printed tolerance is for.

Now the attribution, which is where the interesting mistake lives. BCF differs from Step 0's GBM T-learner in *three* ways at once — (i) a sum-of-trees **posterior** instead of a single point fit, (ii) a **shrink-toward-homogeneity prior** on the treatment surface, (iii) the **propensity** $\hat{e}(x)$ fed in as a covariate — so a two-horse race cannot tell you which of the three did the work. The tempting story is that it was (ii) and (iii): *the Bayesian machinery.* The notebook can **test** that, so it does. Depth A fits a **BART T-learner** on this same data: it has (i) and **neither (ii) nor (iii)**. That completes the ladder, and Depth A prints it:

(1) GBM T-learner → (2) BART T-learner prices **the base learner**. **(2) BART T-learner → (3) BCF** prices **BCF's extra machinery** — the prior and the propensity trick.

Depth A prints an `ATTRIBUTION` block that walks those two rungs with the same tolerance. **Read it there; it is computed, not typed, and it is the only place in this notebook where a claim about *why our own estimator won* is allowed to come from.** In the committed run it is unambiguous: rung (1)→(2) is where the gain is, and rung (2)→(3) is inside the noise band or *negative*. Plainly: **what beat the classical arm is the base learner — averaging over a posterior of tree ensembles instead of committing to one point fit — and BCF's own two ingredients earn nothing on top of a plain BART T-learner here.** That is the ordinary Bayesian gain (model averaging beats committing, when the signal is thin at $n = 1{,}600$ against $\sigma \approx €8$ of noise), not the exotic one, and it is exactly what §7's caveat predicts: with the confounders **fully observed** and overlap good, the propensity trick has nothing left to buy. If a future re-run ever prints "BCF" on rung (2)→(3) beyond the tolerance, *that* — and only that — licenses crediting the prior. Until then, do not.

3 · On the intervals, the *average* hides a disaster — and the disaster is where the money is. This is the finding we did not expect and will not bury.

Overall 90% coverage is the least interesting line in the table. Both arms sit in the mid-to-high 80s, both mildly overconfident, and the `cov90` winner line is decided by a gap so close to the tolerance that it has landed on *both* sides of it across two consecutive FULL runs of this identical notebook. **Read the line, believe the line, and read it as "these are the same to within the noise" whichever name it prints.** (If you want a lesson in why this notebook now computes its verdicts rather than typing them: that is it. A hard-coded "coverage is a wash" would have been *false* on one of those two runs — and a hard-coded "BCF wins coverage" would have been false on the other.)

Split by decile of the true effect, though, and the two arms come apart — and this one is not marginal. The bootstrap band around the classical T-learner covers far more of the **largest-effect** customers than BCF's credible interval does; read the `top-dec` row, which clears its tolerance by a mile and has held in that direction across every run we have done. The cause is not a bug: shrinkage toward homogeneity pulls the biggest τ_i **down**, and the posterior then reports a *narrow* interval around the shrunken value. The classical arm, having no shrinkage to apply, is less accurate on average and more honest at the extreme. **The model is most confident, and most wrong, exactly about the customers a targeting rule is about to spend money on.**

And now put that next to §2's ladder, because together they are the sharpest thing in this section. The comparison above races BCF's *posterior* against a *bootstrap band* — different objects, so it can only take you so far. Depth A therefore runs the clean version: **BART T-learner vs BCF, same base learner, one ingredient apart**, and prints their top-decile coverage side by side. Read that line. What it shows is a prior that, on this data, buys **nothing** on PEHE or AUUC over the plain BART T-learner (§2's ladder) while its shrinkage is exactly the mechanism that flattens the customers we most want to be right about. **A prior is a claim about the world, and on this world it is close to all cost and no benefit — and we know that because we measured it, not because it sounds humble.** That is what Depth A's coverage caveat and the closing paragraph have always gestured at; what they lacked, and now have, is a *comparator* and a tolerance.

4 · On the decision, the posterior bought almost nothing — and the reason to prefer

it is not the euros. Read the profit table. Emailing everyone *destroys* value (the sleeping dogs see to that), and every model-driven rule recovers the great majority of the oracle's profit. But now look at the top two rows of it: §6's honest posterior rule, $P(\tau_i > \text{cost}) > 0.8$, and the classical **bootstrap-exceedance-share** rule at the identical 0.8 threshold finish *level* — the printed gap between them is a rounding error against a five-figure prize — and both beat targeting on either arm's bare *point* estimate (which over-contacts, because a point estimate cannot tell a confident €9 from a coin-flip €9). **On this decision, at this n, the Bayesian layer bought essentially no extra euros. Say it out loud.** So what is the durable argument?

It is the last table, and it is not a philosopher's quibble. Both arms emit a number in $[0, 1]$ per customer, and the two numbers **answer different questions**. $P(\tau_i > \text{cost})$ is a probability about *this customer's effect*, computed by integrating a posterior — the object §6's rule, §6's confidence sweep, Depth C's **straddler** set and Depth C's euro **value of information** are every one of them defined as expectations over. The bootstrap share is the frequency with which a *refitted estimator* would exceed €8 across resampled customer lists: it is centred on $\hat{\tau}(x)$, it inherits the learner's shrinkage in every single replicate (so it cannot warn you that it is biased — the top decile is the proof), and nothing licenses reading it as a degree of belief about the customer. That it *performs* nearly as well here as a decision heuristic is the honest, uncomfortable finding, and we report it. What it cannot do is be *interpreted*, *dialled* against a stated risk appetite, or fed to a VOI integral without silently changing the meaning of every euro that comes out.

What Bayes did NOT buy here. Not the ATE — Step 0 had it, three ways, with honest HC1 and bootstrap intervals. Not immunity from the untestable assumption: Depth B's confounder drags the posterior exactly as far as it drags OLS. Not calibrated intervals on the biggest effects — it is *worse* there than the bootstrap, which is why the closing recommendation is a **test on the straddlers**, not blind faith in $\hat{\tau}$. Not, on this data, a large pile of extra euros. And — per §2's ladder — **not a vindication of BCF's prior or of the propensity trick**, neither of which beats a plain BART T-learner here. What it bought is (a) a **better estimate of** $\tau(x)$ than the classical point fit, earned by the *base learner* (PEHE, AUUC — measured, and never claimed to be seed-stable, because BART is not), and (b) the **decision quantity itself**: a per-customer probability that a CMO can threshold, sweep and price. On a smaller list, or a noisier outcome, the averaging margin grows. On this one, be honest: the classical arm got remarkably close, and the *specifically* Bayesian-causal machinery got nothing.

6 · Decide, in euros

An effect estimate is useless until it becomes an action and a number a manager signs off on. Three moves:

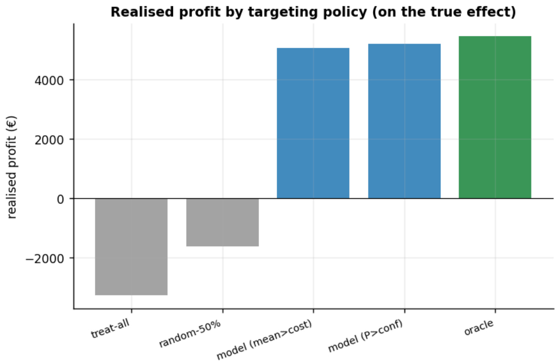
Compare targeting policies on realised profit. We evaluate five rules on the *known truth* (only possible in simulation) so we can see, in €, how much targeting beats the alternatives and how close it gets to the unbeatable oracle: - *treat-all* (email everyone), *random-50%*, - *model (mean > cost)* — email where the posterior-mean effect beats the €8 cost, - *model (P > conf)* — the **honest rule**: email only where we're *confident* the effect beats cost (see below), and - *oracle* — email exactly the customers whose *true* effect beats cost (the best any model could do).

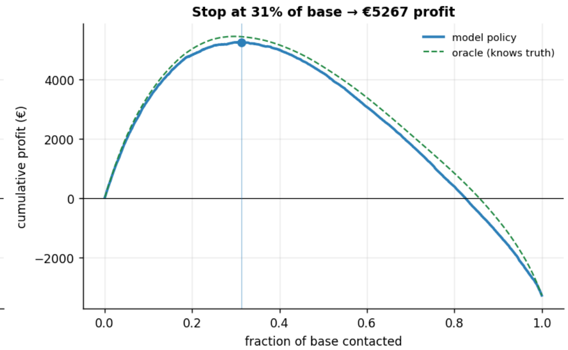
The honest targeting rule. We do **not** target on "mean effect > cost." A customer whose point estimate says "target" but whose interval straddles the cost line is a coin-flip, not a yes. So we target where $P(\tau(x) > \text{cost})$ is **high** (here > 0.8) — probability, from the posterior, that the effect really clears the cost.

The profit curve. Rank customers by estimated effect, contact down the list, and plot cumulative (`true effect − cost`). It rises while we're adding profitable customers and falls once we hit unprofitable ones; its **peak is the optimal stopping point** — "email the top X%, make €Y."

policy	profit	frac_contacted	profit_vs_all
treat-all	-3277	100%	0
random-50%	-1633	49%	1644
model (mean>cost)	5058	39%	8335
model (P>conf)	5210	28%	8486
oracle	5459	30%	8735

```
Email everyone: €-3,277  *  honest model rule: €5,210 (95% of the oracle's
profit), targeting 444 of 1600 customers.
```





6b · The same pipeline on real data (Hillstrom email experiment)

Everything above used *simulated* data so we could grade recovery. Here we run the identical estimator on a **real, public dataset you don't have to provide** — it's fetched over the network from a public URL:

> **Hillstrom / MineThatData (2008)** — 64,000 real customers randomly assigned to *Men's email*, *Women's email*, or *no email*, with downstream visits, conversion and spend. We reduce it to a binary *Women's-email vs no-email* comparison. Because assignment was **randomized (a true A/B test)**, the average effect is trustworthy with no adjustment — but there is **no ground-truth per-customer effect** here (real customers only live one future), so we can estimate and decide, but we *cannot* grade CATE recovery. That's exactly what production looks like: the validation lives in the simulation above; the money is made here.

The cell is **gated** — it fetches only if you set `CMP_REAL=1` (so the offline test suite stays deterministic). Set it and re-run to see the real-data version.

> **What you'll see — a recorded run.** The committed notebook keeps the gate off so the repo runs offline, but for reference a recorded run of this cell (FULL profile, `CMP_REAL=1`) gives: the two-arm slice has **n = 42,693** customers (21,387 emailed, 21,306 held out), and the randomized comparison yields a causal **ATE ≈ €0.42 per**

customer (SE ≈ €0.13) — a genuine, significant lift that pays for itself at any realistic cost of sending an email. On the 4,000-customer subsample, BCF flagged roughly one customer in ten as a confident target (P(effect>0) > 0.8), and ranking by predicted uplift put on the order of **€0.9k of observable incremental spend in the top-30% bucket versus essentially nothing (slightly negative — spend is noisy and zero-inflated) for a random 30%** — a real observable-Qini win, computed with no counterfactual. Two different kinds of number in that list: the sample sizes and the ATE are exact properties of the fixed public dataset and reproduce to the cent; the BCF-derived read-outs (the ~10%, the Qini split) are BART-posterior quantities and pymc-bart is not seed-stable across machines, so treat those as orders of magnitude and run `CMP_REAL=1` to get your own.

```
Real-data section skipped. Set CMP_REAL=1 and re-run this cell to fetch the
Hillstrom experiment.
On real data there is no ground-truth tau, so Steps 5's recovery checks do not
apply.
```

7 · Caveats — the honest failure modes

- **Unconfoundedness is untestable.** If a *hidden* driver moved both who got emailed and how much they spend (say a salesperson's hunch we never recorded), the estimate drifts. Depth B quantifies exactly how strong such a confounder would have to be to flip the decision — as an **E-value**.
- **Overlap matters.** Where some kind of customer was *always* emailed, we're extrapolating, not comparing. We checked the tails in Step 4 and can trim them.
- **Estimator choice is not cosmetic.** The S-learner regularises the treatment signal away and under-detects heterogeneity. *Which axis that breaks depends on the base learner*, so read Depth A's printed by-axis verdict rather than a slogan: with BART it is worst on **PEHE**, **dents** (does not wreck) the ranking, and — unambiguously — **collapses the intervals**, while with the GBM of Step 0 it costs the ranking essentially nothing. A uniformly flattened $\tau(x)$ can still put customers in the right *order*; what flattening always destroys is *magnitudes*, and what it destroys here is *honesty*.
- **BCF's extra machinery earns nothing here — and we measured that, rather than asserting it.** The propensity trick earns its keep specifically under *targeted selection*; don't oversell it as a free lunch. 5x does find BCF beating the *classical* GBM T-learner on PEHE and AUUC — but Depth A's ladder shows that win is a **base-learner** win (a sum-of-trees posterior beating a single point fit), because a plain **BART T-learner** — posterior averaging, but *no* shrink-to-homogeneity prior and *no* propensity input — matches or beats BCF on this data. Read the printed `ATTRIBUTION` block, not this bullet. With confounders fully observed and overlap good, there is simply nothing left for $\hat{e}(x)$ to fix.
- **Our per-customer intervals are worst exactly where the money is.** 5x grades BCF's credible intervals against a bootstrapped classical T-learner and BCF **loses on the top decile of true effect**: the shrink-toward-homogeneity prior pulls the biggest effects down and then reports a *narrow* interval around the shrunken value. Note the sting, which the bullet above sets up — that prior is the ingredient Depth A shows earning *nothing* on PEHE or AUUC over a plain BART T-learner. Here, it is close to **all cost and no benefit**. Read every `P(τ>cost)` as a decision *input*, not gospel, and note that this is precisely why the recommendation ends on "test the straddlers".

- **Sleeping dogs are real.** Negative-effect customers mean "email everyone" can be *value-destroying*, not merely wasteful — which is exactly what the policy comparison in Step 6 showed.

Depth A · Estimator bake-off & failure modes

Why a whole section on picking the estimator? Because the choice is not cosmetic — a popular default (the S-learner) systematically **flattens** heterogeneity and would quietly wreck the targeting. We line up three estimators against the known truth, in two worlds, on the metrics that actually matter for targeting.

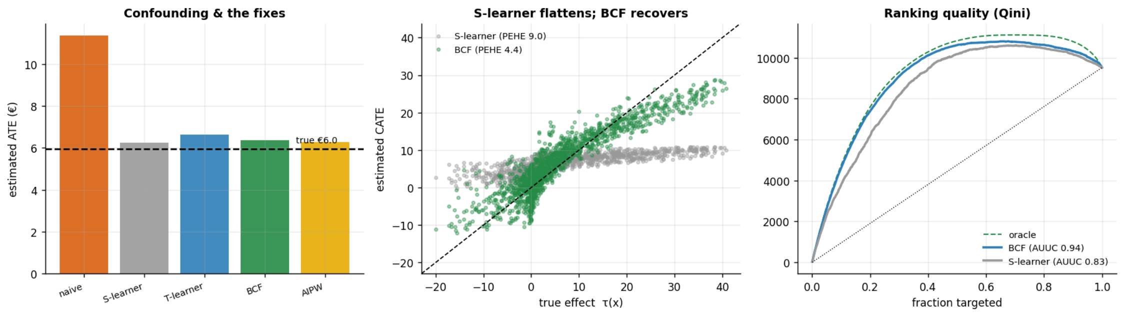
The three estimators (meta-learners — recipes that turn any regression into a CATE estimator): - **S-learner** ("S" = single) — fit *one* model of spend on features *and* treatment together, then read the effect as (prediction with T=1) − (prediction with T=0). Simple, but the model can shrink the treatment's influence toward zero, **flattening** real differences. This is the labelled failure mode. - **T-learner** ("T" = two) — fit *separate* models on the emailed and not-emailed customers, then subtract. A solid default when treatment is randomized. - **BCF** — the propensity-aware, shrink-to-homogeneity model from Step 4; best under confounding.

The two worlds: - **Randomized** — treatment was a coin flip, so *unconfoundedness holds by design* and even simple methods are unbiased on the average. - **Observational** — the confounded world we've been in; here the naive "emailed − not-emailed" average is badly biased, and the estimator has to actively undo the confounding.

The metrics: PEHE (magnitude error, €), corr (ranking correlation), **AUUC** (ranking quality vs oracle), ATE_bias (average-effect error, €), and cov90 (are the 90% intervals honest?).

```
Naive 'emailed - not emailed' (observational, Step 0): €11.4 vs true €6.0 ->
bias +5.4  (confounding nearly doubles the apparent effect)

          regime  estimator  PEHE  corr  AUUC  ATE_bias  cov90
0     randomized  S-learner  9.39  0.75  0.84     -1.09   0.38
1     randomized  T-learner  4.19  0.93  0.96     -0.54   0.88
2     randomized        BCF  4.60  0.91  0.95     -0.45   0.85
3  observational  S-learner  9.01  0.75  0.83      0.31   0.41
4  observational  T-learner  4.22  0.92  0.94      0.70   0.92
5  observational        BCF  4.91  0.90  0.93      0.38   0.88


THE LADDER (observational - this notebook's own world), one ingredient at a
time:
  fit                                   PEHE   AUUC   cov90
  (1) GBM T-learner   [point fit]       5.21   0.89    0.86 *
  (2) BART T-learner  [posterior]       4.22   0.94    0.92
  (3) BCF = (2) + shrink prior + e(x)   4.91   0.93    0.88
  * (1)'s interval is a BOOTSTRAP band around the estimator, not a posterior -
it is a
    different object, so we compare it on PEHE/AUUC only and never on coverage.
```

```
  RUNG (1)->(2)  what the BASE LEARNER buys (GBM point fit -> BART posterior;
nothing else):
      PEHE   BART-T            (5.21 vs 4.22;  |gap| 0.98 >  2xSE = 0.21)
      AUUC   BART-T            (0.894 vs 0.936;  |gap| 0.042 >  2xSE = 0.009)

  RUNG (2)->(3)  what BCF's EXTRA MACHINERY buys (shrink-to-homogeneity prior +
e(x)):
      PEHE   BART-T            (4.22 vs 4.91;  |gap| 0.68 >  2xSE = 0.15)
      AUUC   BART-T            (0.936 vs 0.928;  |gap| 0.008 >  2xSE = 0.003)
      cov90  too close to call (0.017 vs 0.024;  |gap| 0.006 <= 2xSE = 0.027)
[distance from the nominal 0.90: 0.92 vs 0.88]
      top-dec BART-T            (0.75 vs 0.30;  |gap| 0.45 >  2xSE = 0.08)
[coverage on the TOP true-effect decile]

  ==> ATTRIBUTION (computed, not typed - every prose cell in this notebook
defers to THIS):
      * the BASE LEARNER (point fit -> posterior) buys: PEHE, AUUC
      * BCF's EXTRA machinery (shrink prior + e(x)) buys: NOTHING beyond the
noise band, on any axis above
      * …and it COSTS: PEHE, AUUC, top-decile coverage
      => so 5x's Bayesian edge over the classical arm is a BASE-LEARNER win - a
sum-of-trees POSTERIOR beating a single point fit, and BCF's own prior and
propensity trick add nothing measurable on top of a plain BART T-learner.
         (Fully observed confounders + good overlap: exactly the regime where §7
says the
          propensity trick has little to earn. This is that caveat, MEASURED.)


  THE S-LEARNER FAILURE MODE, axis by axis (BART S vs BART T - same base
learner, so this
  isolates the 'one shared surface' choice):
      PEHE   BART-T            (9.01 vs 4.22;  |gap| 4.78 >  2xSE = 0.40)
      AUUC   BART-T            (0.832 vs 0.936;  |gap| 0.104 >  2xSE = 0.014)
      cov90  BART-T            (0.493 vs 0.017;  |gap| 0.476 >  2xSE = 0.031)
[distance from the nominal 0.90: 0.41 vs 0.92]
      => the S-learner still captures 83% of the oracle's ranking value (random
= 0%),
         so on THIS axis the flattening is a DENT, not a wreck - and with a GBM
base learner
         (Step 0) it cost the ranking nothing at all. The unambiguous casualty
is the
         INTERVALS: 41% coverage against a nominal 90%. Flattening tau(x) hurts
magnitudes
         always, ranking sometimes, and honesty every time.
```



Read-out. Everything below is *read off the printed blocks above*, and every head-to-head there is judged against the same tolerance — twice the paired-bootstrap SE of the gap. That matters more than usual here: `pymc-bart` is **not seed-reproducible** in this environment, so a small BART gap is not a result, *whichever direction it points.* This section and 5x used to apply two different standards to two different gaps (dismissing a larger one as noise, banking a smaller one as skill). Now they both defer to the same computed verdict.

(1) The S-learner is the clear loser — but be precise about *what* it loses. Its single shared surface regularises the treatment signal toward zero, so it **flattens** heterogeneity even with no confounding to fight, and the printed by-axis verdict shows the damage is not uniform: worst on **PEHE**, a genuine but survivable **dent** to the ranking (it still captures a large share of the oracle's AUUC — the print says how much), and an outright **collapse of the intervals** (`cov90`, the unambiguous casualty). With Step 0's GBM base learner the same S-recipe costs the ranking essentially *nothing*, which is the tell: the severity is a property of the base learner, not of the letter "S". Tellingly, the **average effect survives**: read the `ATE_bias` column and compare it to the naive difference-in-means printed at the top of this section (off by about **+€5** against a ~€6 effect). All three estimators — the S-learner included — land an order of magnitude closer than that; *which* of the three has the smallest ATE bias shuffles from run to run and is not a finding. So flattening hurts per-customer effects, their ranking and their intervals, **not the mean** — which is exactly why a response-style average would sail straight past the problem and only the heterogeneity metrics expose it. **The S-learner's fault is heterogeneity, not the average, and an ATE-only dashboard would have shipped it.**

(2) The ladder, and what it says about the Bayesian win in 5x. The bake-off exists to answer a question 5x cannot: BCF beat the classical GBM T-learner there, but BCF changes *three* things at once (a posterior instead of a point fit, a shrink-to-homogeneity prior, a propensity input). The **BART T-learner** has the first and neither of the other two, so it splits the credit — and the printed `ATTRIBUTION` block does the split, with the tolerance applied. **Read its verdict there, not here.** In the committed run it says the base learner is what bought the gain, and BCF's own machinery adds nothing measurable on top of a plain BART T-learner — on any axis printed. That is not a defeat for the Bayesian arm; it is the *correct* Bayesian story, and it is precisely what Step 7's caveat predicts: with the confounders fully observed and overlap good, feeding $\hat{e}(x)$ in has nothing left to fix. BCF stays the primary model here because its propensity input earns its keep under **targeted selection** — the regime Depth B stress-tests, and the one it was built for — not because this table crowns it. All three (plus the methodologically different, doubly-robust AIPW) still agree on the **ATE**, which remains our cross-method credibility check.

Coverage — the honest caveat we don't hide. BCF's 90% intervals tend to **under-cover on the largest true effects** — read the `90% coverage` printed above (and the bake-off `cov90` column) for *this* run, and see the coverage-by-decile panel in Step 5, where the top true-effect decile is worst. The exact figure **wobbles run-to-run** because pymc-bart's sampler is not seed-reproducible; the T-learner's intervals are usually a touch better calibrated, and the S-learner under-covers badly (the labelled failure-mode). The takeaway is deliberately conservative: heterogeneity is hardest to pin exactly where the effect is biggest, so we treat every `P(τ>cost)` downstream as a decision *input*, not gospel, and keep the "test the straddlers" recommendation — it hedges precisely where calibration is thinnest.

Depth B · Identification rigour & sensitivity

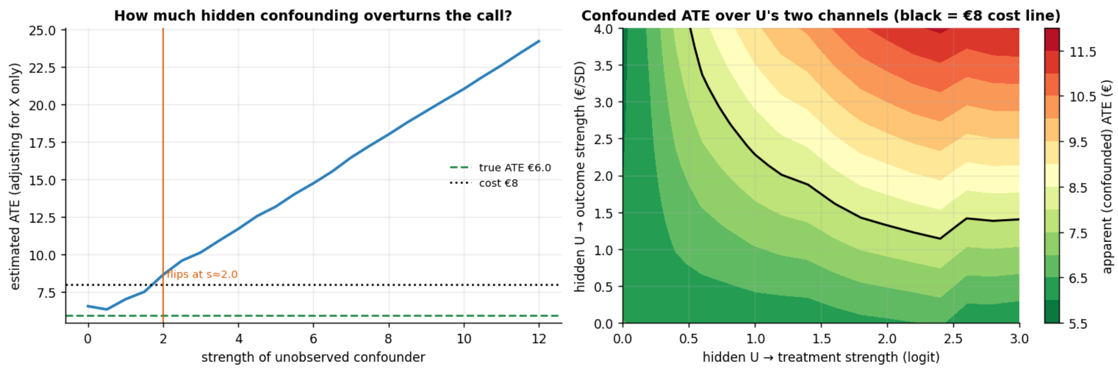
We already confirmed **overlap** (Step 4). The deeper worry is **unconfoundedness**, which is *untestable* — so the honest question is not "is it true?" but **"how fragile is our conclusion to a confounder we didn't measure?"**

We simulate a hidden driver U that moves *both* who gets emailed and how much they spend, sweep its strength, and refit adjusting for the observed features **only** (blind to U, as we'd be in reality). We report two complementary things a CMO can act on:

- the **tipping point** — the confounder strength at which the (wrongly) adjusted effect crosses the €8 cost line and would flip us from "target selectively" to "email everyone pays" (a *decision*-flip number);
- the **E-value** (VanderWeele-Ding) — the minimum strength of association a hidden confounder would need with *both* treatment and outcome, on a **risk-ratio scale** — a risk ratio being how many times more likely an outcome is under one condition than another, i.e. a multiplicative scale — to explain the *positive effect* away entirely (to zero). The E-value is defined only on the risk-ratio scale, so we standardize the euro effect by the outcome SD (VanderWeele's continuous-outcome rule) and compute it on the **AIPW estimate** — never on the simulation truth, which a real analyst never has. A large E-value means "you'd need an implausibly strong hidden confounder to overturn this"; a modest one (as we find here) means the robustness is real but not decisive.

The right panel makes the tipping point two-dimensional: it is a **genuinely refit** surface of the confounded ATE as U's two channels (into treatment, into outcome) vary independently — so the black cost line traces the real (a, b) frontier, not a drawn-in gradient — omitted-variable bias (**OVB**) in two dimensions, where the bias scales with *both* channels, $a \times b$.

```
Tipping point: the observed-feature-adjusted ATE crosses the €8 cost line at
hidden-confounder
strength ~ 2.0 - beyond that, confounding alone could fake an 'email everyone
pays' signal.
E-value 2.07 (at the near CI limit, 1.94): to explain the positive €6.28 effect
away to zero, a hidden confounder would need ~2.1x association (risk-ratio
scale) with BOTH email
and spend. That is moderate, not decisive - ask a domain expert whether anything
that strong went unmeasured.
```

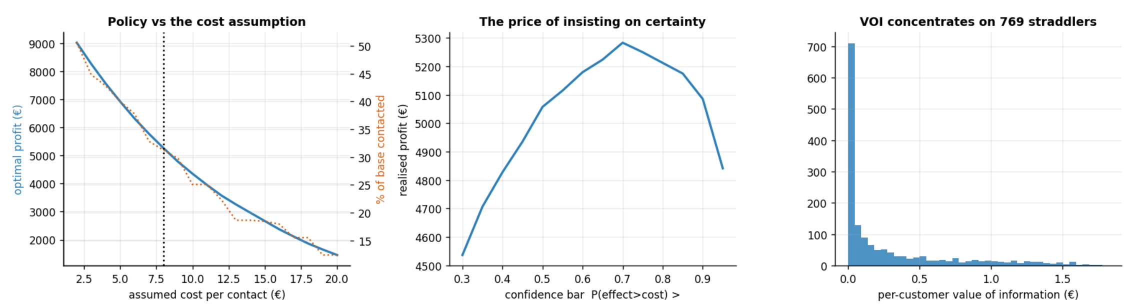


Depth C · Euro policy — sweeps, value of information & test design

The decision is a number *under assumptions*. We stress the two a manager will argue about — the per-contact **cost** and how **confident** we insist on being — and then price the remaining uncertainty and turn it into a concrete experiment.

Value of information (VOI). Uncertainty has a *price*: the euros we lose by acting on the noisy estimate instead of the (unknown) truth. It concentrates on the **straddlers** — customers whose credible interval crosses the cost line, so we genuinely can't tell if they're worth contacting. The total VOI on the straddlers is an **upper bound** on what any experiment on them can be worth: it prices *perfect* information, whereas a finite A/B test resolves only part of the uncertainty and so is worth strictly less. Still, it sets the ceiling on sensible test spend — and tells you on whom.

```
Total value of information €467; €452 of it sits on the 769 'straddler'
customers (48% of the base) whose interval crosses the cost line.
```



The one-paragraph decision

> Targeting the model's *confident* persuadables makes materially more money than emailing everyone (which here **loses** money to sleeping dogs), and captures most of the oracle's profit. The call is robust to plausible cost assumptions; hidden confounding would

have to be *moderately* strong to overturn it — the confounded ATE only crosses the cost line at confounder strength ≈ 2.0, and the positive effect carries an E-value ≈ 2 (real but not decisive robustness, so this is a finding to keep watching, not to bank). The uncertainty that remains is worth real money and is concentrated on the "straddler" customers whose interval crosses the cost line — **so the follow-up is a small randomized A/B test on the straddlers, not a blanket send.** Act where we're sure; measure where we're not. *(One honest caveat: BCF's per-customer intervals undercover the largest individual effects — the coverage-by-decile diagnostic above shows the top decile falling short of nominal — so the straddler set is a conservative upper bound, and the biggest-effect customers' $\hat{\tau}$ should be read as a floor, not a point.)*


```
{"true_ate": 5.952, "bcf_ate": 6.374, "aipw_ate": 6.283, "PEHE": 4.445,
"AUUC": 0.9413, "coverage90": 0.8844, "profit_treat_all": -3277.0,
"profit_model": 5210.0, "profit_oracle": 5459.0, "sensitivity_tip": 2.0,
"e_value": 2.07, "VOI_total": 467.3, "n_straddlers_to_test": 769}
```